

EN 601.124

The Ethics of Artificial Intelligence and Automation:
Human Subjects and Human Data

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Announcements

- First reading reflection is due **tomorrow EOD** (submission via Gradescope)
- For this, need to read 2 articles and answer a few questions
- Graded for submission only.

The assignment: Submit a short reflection. Answer each question, in up to 200 words for each question:

1. What is the goal of the paper in your own words?
2. What are the paper's main strengths, technically, empirically, or conceptually? What did you like the most, or why do you think it is important?
3. What ethical risks or harms does the paper have?
4. What concrete changes would be required to make this work ethically stronger if it were revised or done again?

Today's lecture

- Last week:
 - Background on AI narratives, history, case-studies of ethical risks
- Today
 - The role of human in AI
 - **Human subjects:** Origins of ethical issues in human research, modern standards and rules for research with humans
 - **Human data:** The role of human data and human research in AI

The role of humans

- AI systems are trained **on people** and **by people**, to be **used by people**
- Ethical failures in AI often mirror earlier failures in human-subject research
- Today's harms are often distributed, invisible, and scaled



Lessons from the history: Medicine

Nuremberg Code of 1947

- World War II medical experiments on prisoners in Nazi concentration camps
- After the war ended, a series of trials were held against major war criminals
- The first trial in 1947 – The Doctors’ Trial
 - “The United States of America v. Karl Brandt, et al.,”
 - 23 physicians from the German Nazi Party were tried for crimes against humanity for murder and torture in the atrocious experiments they carried out on unwilling prisoners of war
 - 16 were found guilty, of which 7 received death sentences and 9 received prison sentences ranging from 10 years to life imprisonment
- The verdict also resulted in the creation of the Nuremberg code a set of 10 ethical principles for human experimentation

Nuremberg Code of 1947

1. Voluntary consent is essential
2. The results of any experiment must be for the greater good of society
3. Human experiments should be based on previous animal experimentation
4. Experiments should be conducted by avoiding physical/mental suffering and injury
5. No experiments should be conducted if it is believed to cause death/disability
6. The risks should never exceed the benefits
7. Adequate facilities should be used to protect subjects
8. Experiments should be conducted only by qualified scientists
9. Subjects should be able to end their participation at any time
10. The scientist in charge must be prepared to terminate the experiment when injury, disability, or death is likely to occur

United States Public Health Service (USPHS) Syphilis Study at Tuskegee

- Conducted by the U.S. Public Health Service from 1932 to 1972 in Macon County, Alabama
- Enrolled 600 Black men (399 with syphilis, 201 without) under the false premise of receiving free medical care
- Participants were not informed of their diagnosis and were never asked for informed consent
- Even after penicillin became the standard cure in the 1940s, treatment was deliberately withheld
- The study resulted in preventable suffering, deaths, and transmission to families
- Public revelation in 1972 led to the study's termination and major reforms in human subjects protections



“I am utterly astounded by the fact that physicians allow patients with a potentially fatal disease to remain untreated when effective therapy is available,” the 1965 letter read. “I assume you feel that the information which is extracted from observation of this untreated group is worth their sacrifice. If this is the case, then I suggest the United States Public Health Service and those physicians associated with it in this study need to re-evaluate their moral judgments in this regard.”

The Washington Post
Democracy Dies in Darkness

Morning Mix

Dr. Irwin Schatz, the first, lonely voice against infamous Tuskegee study, dies at 83

April 20, 2015 More than 10 years ago



Make us preferred on Google



Lessons from the history: Behavioral studies

Stanford prison experiment

- Conducted in **1971** by psychologist Philip Zimbardo at Stanford University
- Simulated prison in the basement of the psychology building
- 24 male college students randomly assigned to roles of “guards” or “prisoners”; terminated after six days
- **Guards** quickly adopted authoritarian and abusive behaviors
- **Prisoners** showed signs of stress, emotional breakdown, and helplessness
- Zimbardo acted as prison superintendent, blurring researcher–participant roles

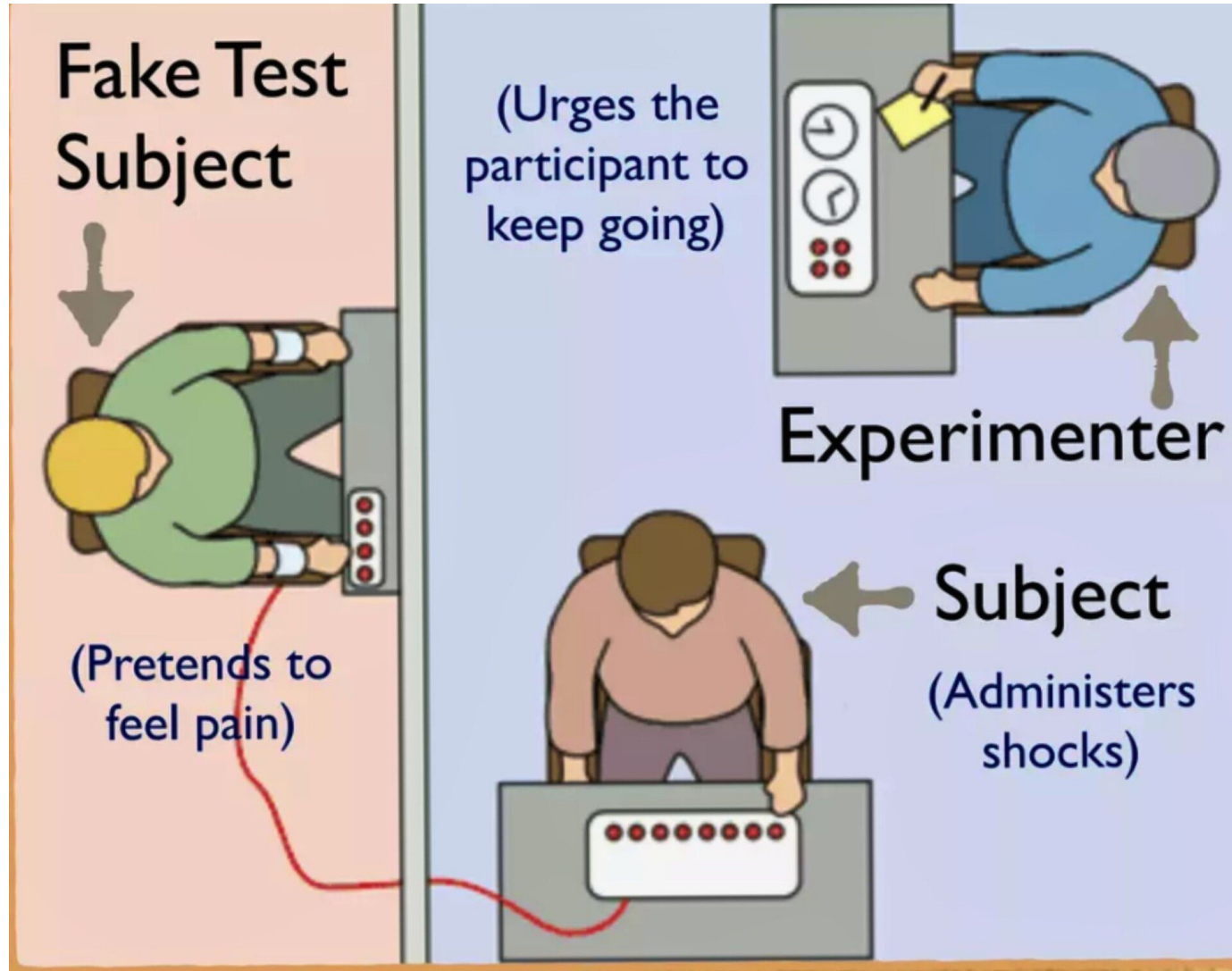


Stanford prison experiment

- **Inadequate informed consent** about risks and potential harms
- **Lack of independent oversight** or stopping rules as harm escalated
- Researcher **role conflict** compromised participant safety
- Participants experienced **psychological harm** without adequate protection
- Findings were later **criticized for demand characteristics and scripting**



Stanley Milgram's Shock Experiments



Stanley Milgram's Shock Experiments

- Conducted in the early **1960s** by psychologist Stanley Milgram at Yale University
- Designed to study obedience to authority, especially in the aftermath of WWII
- 65% of participants administered the maximum shock level when instructed
- Subjects showed visible distress, yet were encouraged to continue by the experimenter
- Raised enduring ethical concerns about deception, psychological harm, and consent

Stanley Milgram's Shock Experiments



- **Claimed values:**

- Demonstrated the powerful role of authority and situational pressure in shaping human behavior
- Provided an influential framework for understanding obedience in bureaucratic and institutional settings

- **Critiques:**

- Heavy use of deception undermined informed consent and participant autonomy
- Caused significant psychological distress without clear necessity
- Participants may have suspected the shocks were fake, raising validity concerns
- Findings are often overgeneralized beyond the experimental context
- Enduring tension: The study's impact on theory came at a high ethical cost, prompting debate about what knowledge justifies harm

Group activity: Would you participate?

- Split into groups
- Discuss ~10 min:
 - Introduce yourself + state a book or a movie you liked recently
 - Under what conditions (if any) would you be willing to participate in these two studies, as originally conducted?
 - Be prepared to share one insight with the class

Stanford Prison Experiment

- **Social roles and institutional contexts** can rapidly shape behavior, overriding individual personality

Milgram Obedience Experiments

- **Ordinary people often obey authority** even when actions conflict with their moral values
 - > Ordinary individuals may engage in **abusive or dehumanizing actions** when given power

Modern standards and rules for human research

National Research Act 1974

These experiments led to the creation of the National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research

- The Common Rule: Title 45, Part 46 of the Code of Federal Regulations: Protection of Human Subjects.
- Required institutional review of all federally funded experiments
- Institutional Review Boards (IRBs)
- Issued Belmont Report in 1976/1979

The Belmont Report: Basic ethical principles

Respect for persons

- Individuals should be treated as autonomous agents
- Participants must give informed, voluntary consent
- Consent requires understanding risks, benefits, and purpose
- Special protections are required for vulnerable populations
- Deception undermines respect unless strictly justified and minimized
- Participants must retain the right to withdraw at any time
- Autonomy is compromised by pressure, authority, or undue incentives

Informed consent



The Belmont Report: Basic ethical principles

Beneficence

- Researchers must maximize possible benefits and minimize possible harms
- Harm includes psychological, social, and dignitary harms (not just physical)
- Risks should be reasonable in relation to expected benefits
- Ongoing monitoring is required to stop studies when harm escalates
- Poor study design weakens ethical justification for exposing participants to risk
- Benefits should be realistic, not speculative
- “No harm intended” is not sufficient if harm is foreseeable

Minimizing harm

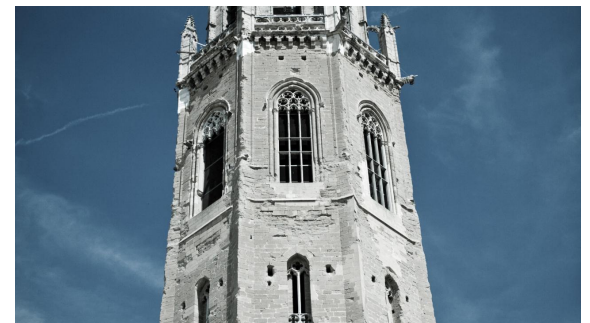


The Belmont Report: Basic ethical principles

Justice

- The burdens and benefits of research should be fairly distributed
- Vulnerable or marginalized groups should not be targeted for convenience
- Those who bear risks should have a reasonable chance to benefit
- Justice extends beyond individuals to communities and populations

research should not
be conducted only for
the ivory tower

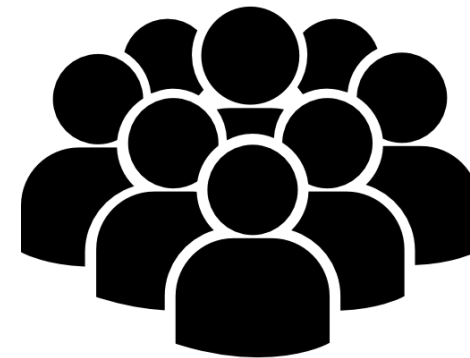


IRB - Institutional Review Board

- An **Institutional Review Board (IRB)** is an independent committee that reviews research involving human subjects
- Its role is to **protect participants**, not to evaluate scientific prestige or novelty
- IRBs assess **risk vs. benefit**, informed consent, and participant protections
- Required for most human-subjects research at universities, hospitals, and research institutions
- Reviews whether studies comply with ethical principles (e.g., **Belmont Report**) and federal regulations
- Can require **changes, monitoring, or termination** of studies that pose undue risk
- Designed to prevent harms

Key concepts

- Human subject: a living individual **about whom** an investigator (professional or student) is conducting research:
 - Obtains information through **intervention** or **interaction** with the individual, and uses, studies, or analyzes the information; or
 - Obtains, uses, studies, analyzes, or generates **identifiable private information**



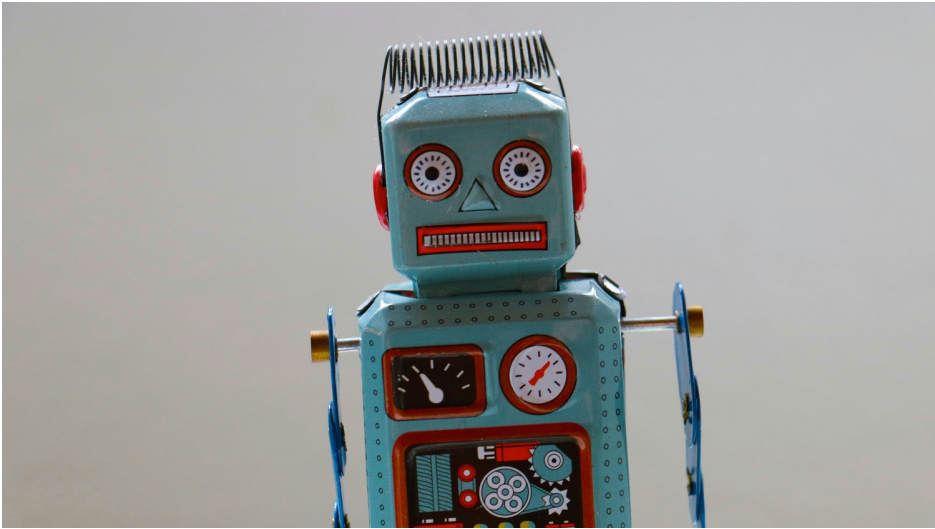
Key concepts

- Disclosure: the extent to which researchers **truthfully inform participants** about the purpose, procedures, risks, and nature of a study.
- The Belmont Report permits limited disclosure or deception only when:
 - The research has **significant scientific value**
 - The deception poses **no more than minimal risk**
 - The research **could not practicably be carried out** without the deception
- When disclosure is withheld or altered, researchers have an obligation to conduct a **debriefing**:
 - Inform participants of the **true purpose and nature** of the study
 - Explain the **reason for the deception**
 - Provide participants the opportunity to **withdraw their data**

Key concepts

- **Exempt research:** Human subjects research qualifies if:
 - Research only includes survey procedures, interview procedures, or observation of public behavior (including visual or auditory recording) and one of:
 - The identity of the human subjects cannot readily be ascertained
 - Any disclosure of the human subjects' responses outside the research would not place the subjects at risk of criminal or civil liability or be damaging to the subjects' financial standing, employability, educational advancement, or reputation
- Secondary research uses of identifiable private information if the identifiable private information is publicly available

Why AI development is often exempt from IRB

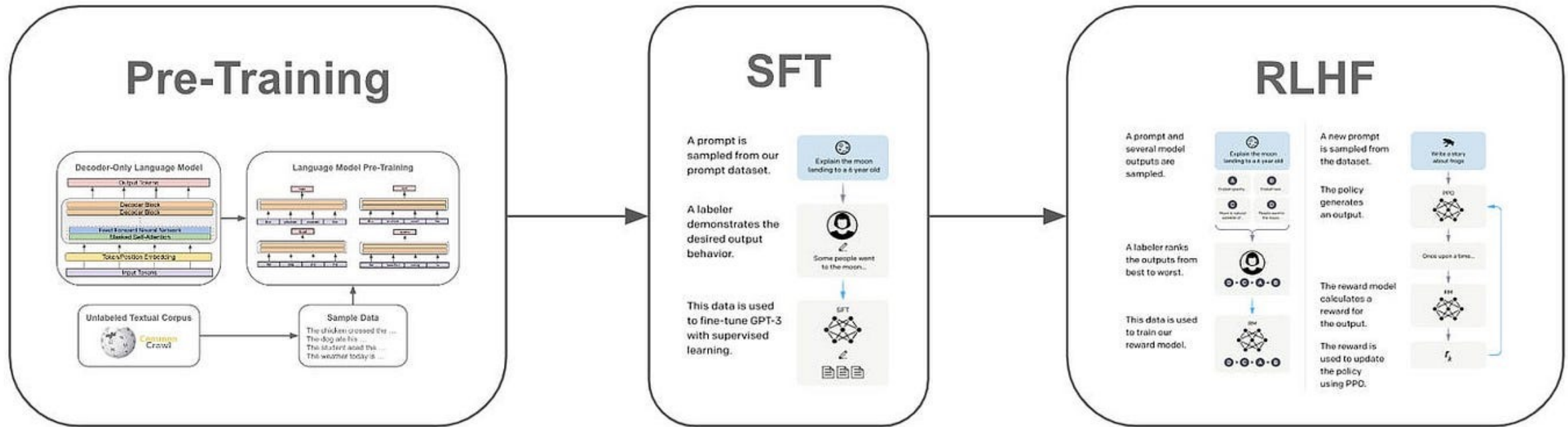


As a result, impactful AI systems can affect millions without formal ethics review!

- IRBs were designed for human subjects research, not software development!
- AI work is often framed as engineering, product testing, or quality improvement
- Use of existing data is frequently classified as exempt
- Online experiments (A/B tests, recommender tuning) may occur outside academic institutions
- Companies are not always subject to the same oversight requirements as universities

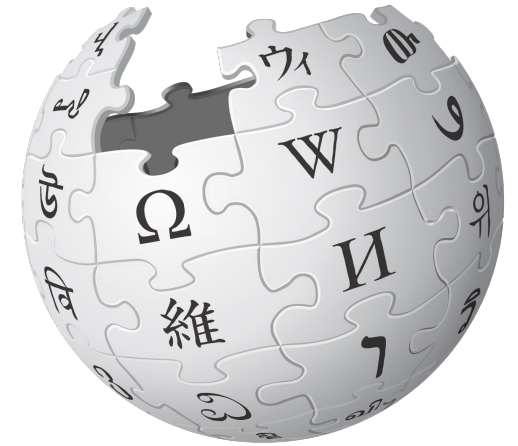
The role of **human** data and **human** research in AI

The case of large language models



Pre-training

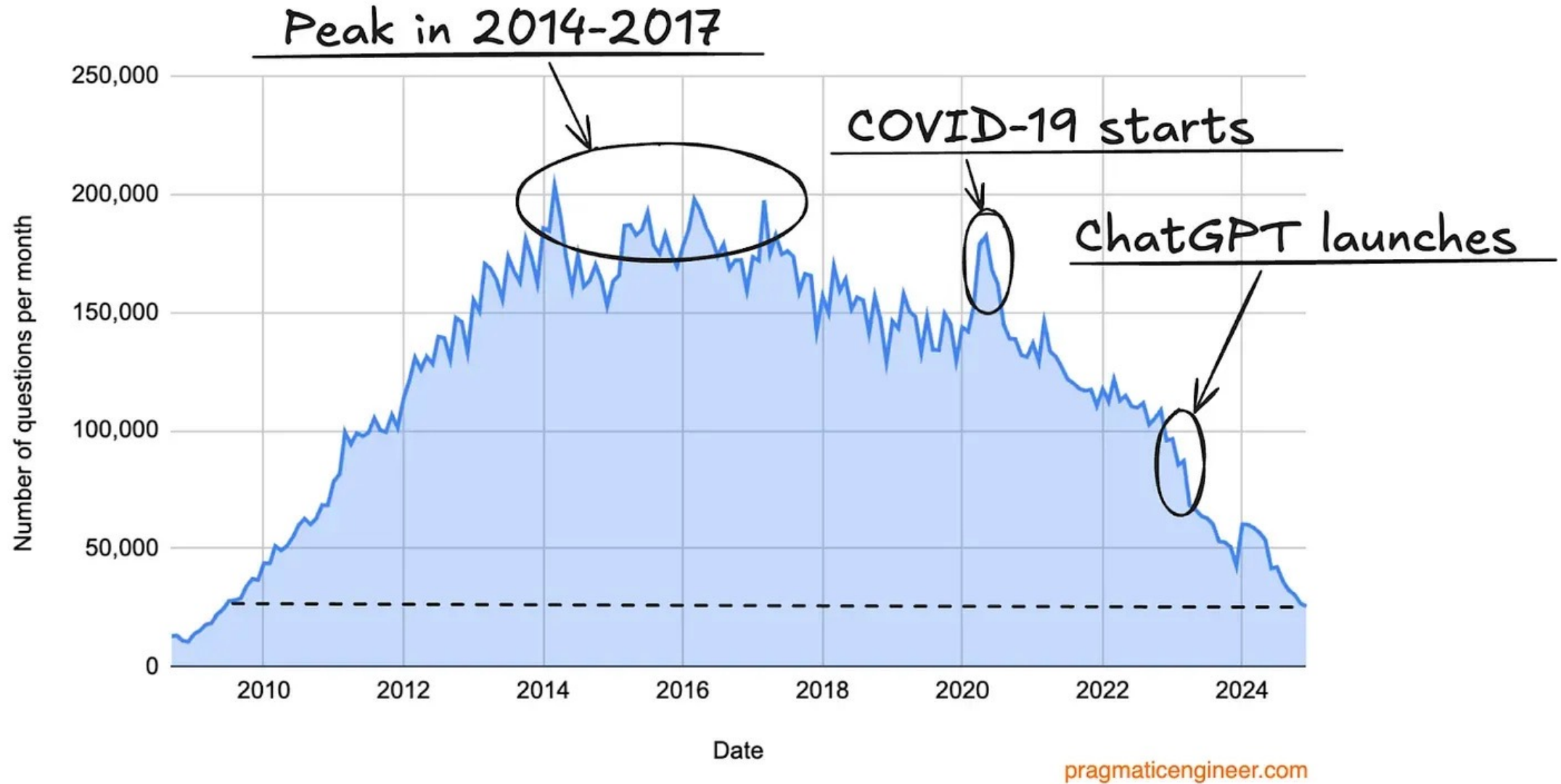
- Input: **Massive corpora of human-written text** (books, articles, websites, code, forums)
- Models learn by predicting the next word/token given prior context
- Output: A language model that can generate new text resembling human writing
- Scale matters: performance improves with more data, parameters, and compute
- High-quality AI requires large amounts of human labor via **past authorship**; contributors do not benefit directly from AI products



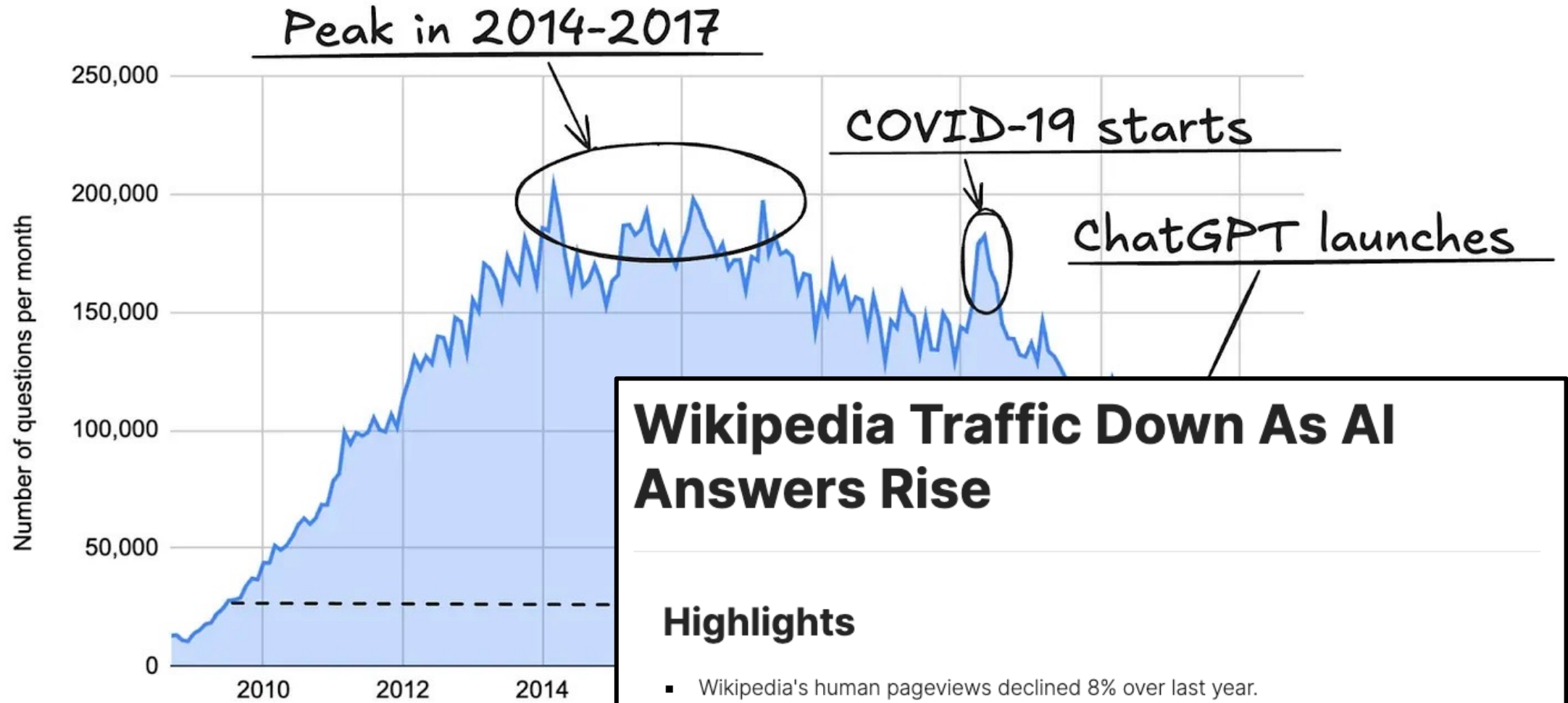
Ethics of using internet data

- Social media data: posts from Twitter, Reddit, YouTube, Telegram, etc.
- From IRB perspective this kind of corpus data is exempt if it is public
 - E.g., public social media data
- But are there still ethical questions:
 - Informed consent
 - Privacy
 - Terms of service
- Consensus: OK to programmatically collect data without explicit consent
 - But seek informed consent for all directly quoted content (in publications, reports)

Monthly questions asked on StackOverflow



Monthly questions asked on StackOverflow



SFT



Good answer ?

- Input: Datasets of **human-written high-quality responses to prompts**
- Data is often created by human annotators (experts or crowd workers)
- Models are trained to imitate desired answers given an input prompt
- Learning objective: minimize error between the model's output and the human-provided target response
- Output: A model that follows instructions more reliably and produces useful responses
- SFT shapes style, tone, and task performance, not raw language ability
- Effective SFT depends on substantial human labor and expertise

RLHF

prompt

Answer
A

or

Answer
B

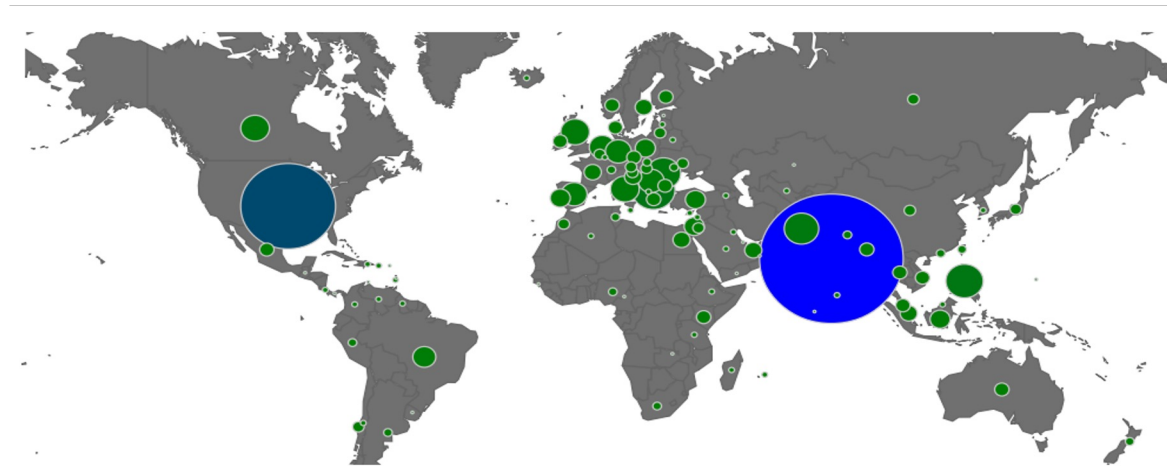
?

- Input: Model-generated outputs evaluated by **human raters (rankings, scores, preferences)**
- Humans judge responses for qualities like helpfulness, safety, accuracy, and tone
- These judgments are used to train a reward model representing human preferences
- Output: A model that better aligns with human expectations and norms
- Like SFT, RLHF relies on large-scale human labor and normative judgment

Ethics of using crowdworkers

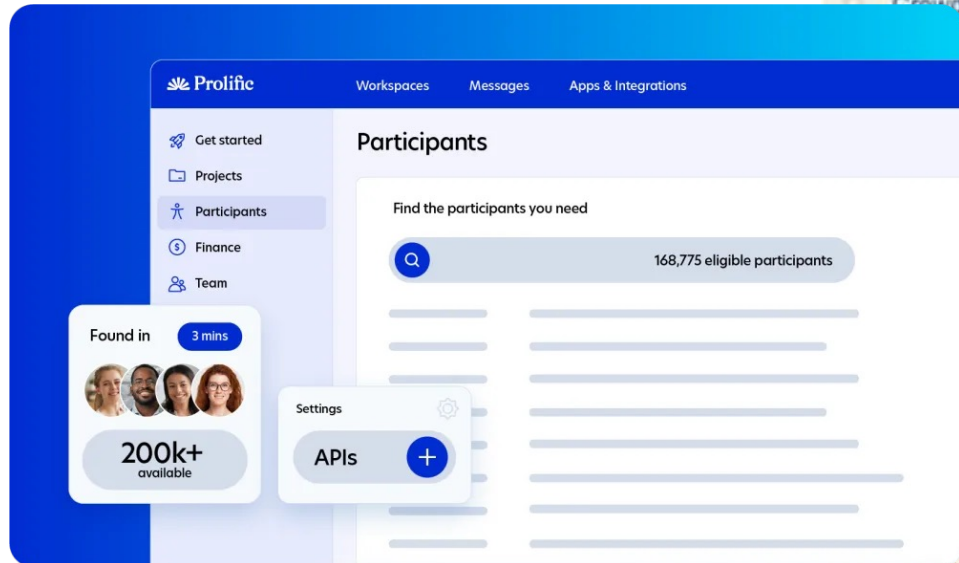
Who are crowdworkers?

- Largely from the US and India
- Skewed young, female, and lower income; many underemployed



Ethics of using crowdworkers

- Amazon Mechanical Turk, Prolific, etc



All HITs Your HITs Queue

HIT Groups (1-20 of 640) [Show Details](#) [Hide Details](#) Items Per Page: 20

Requester	Title	HITs	Reward	Created	Actions	
ScoutIt	Classify Receipt	151	\$0.03	14s ago	Preview	Quality
Crowdsurf Support	Full Text Review - Earn up to \$...	53	\$0.17	3m ago	Preview	Quality
Laura A. King	Personality, Information Proce...	1	\$0.15	4m ago	Preview	Accept & Work
Crowdsurf Support	Review, edit, and score the tra...	1,091	\$0.02	5m ago	Preview	Quality
Fissel	Quick Demographic Survey!(-...	1	\$0.01	6m ago	Preview	Accept & Work
t	Extract summary information fr...	1	\$0.05	9m ago	Preview	Accept & Work
surf Support	Transcribe up to 35 Seconds o...	1,042	\$0.05	10m ago	Preview	Quality
evens	Help Pick a Book Cover!	1	\$0.10	12m ago	Preview	Accept & Work
t	Extract summary information fr...	1	\$0.05	12m ago	Preview	Accept & Work
el Busseri, PhD	Answer survey (10 minutes) a...	1	\$1.00	12m ago	Preview	Quality
linnikin	Feedback Seeking Motives Pr...	111	\$0.25	15m ago	Preview	Quality
SEO BrainTrust	Summarize and write three ke...	23	\$0.35	25m ago	Preview	Quality

Ethics of using crowdworkers: 1) **income**

- Multiple studies have found actual wages at or below \$2 / hr
 - Our task-level analysis revealed that workers earned a median hourly wage of only ~\$2/h, and only 4% earned more than \$7.25/h. While the average requester pays more than \$11/h, lower-paying requesters post much more work. (Hara et al. 2018)
- Need to consider time finding tasks, lost wages from rejections, etc
- **No job security, benefits, collective bargaining, ability to know their employers**
- Frequently secondary source of income, sometimes primary
 - 20%+ use crowdwork income to “to make basic ends meet”

BUSINESS • TECHNOLOGY

Exclusive: OpenAI Used Kenyan Workers on Less Than \$2 Per Hour to Make ChatGPT Less Toxic

15 MINUTE READ



Ethics of using crowdworkers: 2) invisibility

- **Invisibility** facilitated by distance, anonymity, minimal communication, and electronic exchange

Programmatically specify the html

Release annotation tasks

Collect responses and approve/reject

```
import boto3

client = boto3.client('mturk')
```

```
response = client.create_hit(
    MaxAssignments=123,
    AutoApprovalDelayInSeconds=123,
    LifetimeInSeconds=123,
    AssignmentDurationInSeconds=123,
    Reward='string',
    Title='string',
    Keywords='string',
    Description='string',
    Question='string',
    RequesterAnnotation='string',
    QualificationRequirements=[
        {
            'QualificationTypeId': 'string',
            'Comparator': 'LessThan' | 'LessThanOrEqualTo' | 'GreaterThan' | 'GreaterThanOrEqual',
            'IntegerValues': [
                123,
            ],
            'LocaleValues': [
                {
                    'Country': 'string',
                    'Subdivision': 'string'
                },
            ],
        },
    ],
    'RequiredToPreview': True|False,
    'ActionsGuarded': 'Accept' | 'PreviewAndAccept' | 'DiscoverPreviewAndAccept'
),
```

```
response = client.approve_assignment(
    AssignmentId='string',
    RequesterFeedback='string',
    OverrideRejection=True|False
)
```

Select a customizable template to start a new project

Survey
Survey Link
Survey
Vision
Image Classification
Bounding Box
Semantic Segmentation
Instance Segmentation
Polygon
Keypoint
Image Contains
Video Classification
Moderation of an Image
Image Tagging
Image Summarization
Language
Sentiment Analysis
Intent Detection
Collect Utterance
Emotion Detection
Semantic Similarity
Audio Transcription
Conversation Relevance
Document Classification
Translation Quality
Audio Naturalness
Other
Data Collection
Website Collection
Website Classification
Item Equality
Search Relevance
Other

Instructions: Given an image, write a sentence summarizing what it shows.
Use punctuation and don't mention that you're describing an image.



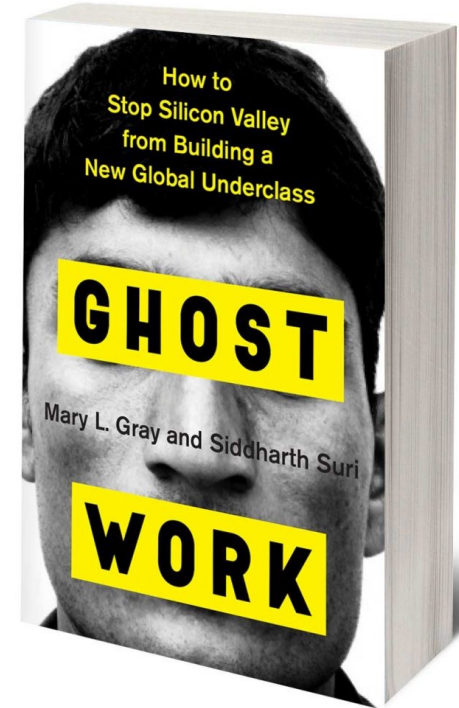
Summarize the image with a sentence...

Submit

Create Project

Ethics of using crowdworkers: 2) invisibility

- **Invisibility** facilitated by distance, anonymity, minimal communication, and electronic exchange
- Crowdworkers appreciate communication and responsiveness, i.e. less invisibility
- Review platforms for crowdworkers to rate requesters



Ethics of using crowdworkers: 3) harms

- Bear the burden of exposure to toxic texts, unsafe images
- No debrief, analysis of benefits and harms (no IRB)

'It's destroyed me completely': Kenyan moderators decry toll of training of AI models

Employees describe the psychological trauma of reading and viewing graphic content, low pay and abrupt dismissals



Office premises of Sama in Nairobi, Kenya. Photograph: Tony Karumba/AFP/Getty Images

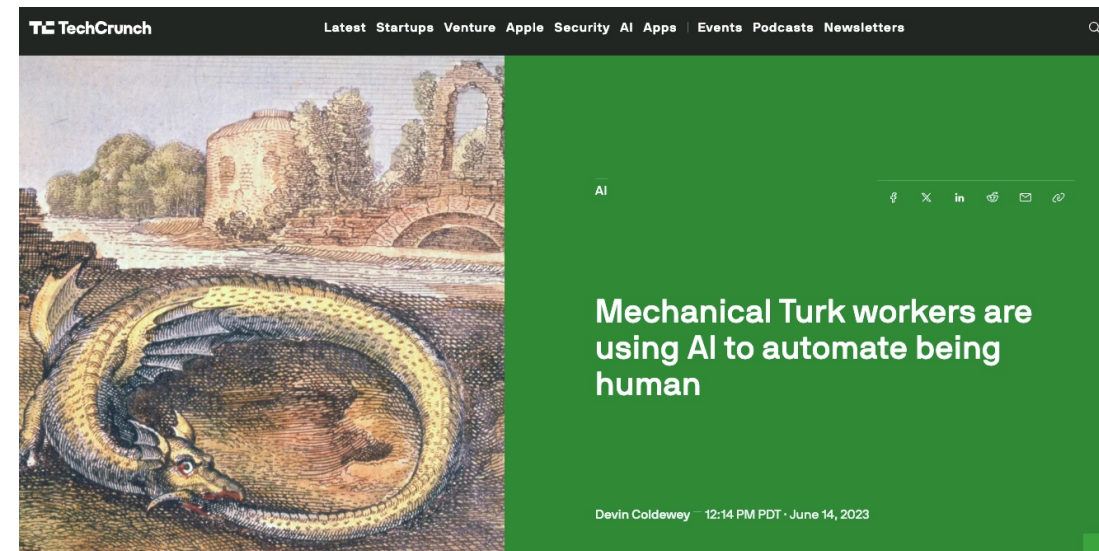
The images pop up in Mophat Okinyi's mind when he's alone, or when he's about to sleep.

Sustainability of human input in AI development

"LLMs are widely used in crowd work. We also find that responses written with the help of LLMs are high-quality but more homogeneous than those written without LLMs' help."

"LLM use by crowd workers compromises research on human behavior, preferences, and opinions. Our results indicate we must find ways to prevent inappropriate LLM use or appropriately incorporate LLMs into crowd workers' workflows."

"LLM use can be diminished by **adding hurdles that complicate their use** or by **asking crowd workers not to use LLMs**. However, LLM use remained common in our field experiments even with these mitigation strategies."



Veselovsky, Veniamin, et al. "Prevalence and prevention of large language model use in crowd work." *Communications of the ACM* 68.3 (2025): 42-47.

Class activity

Split into groups, and discuss:

How should humans involved in AI development be compensated?

1. Who should be paid? (annotators, evaluators, data contributors, affected communities)
2. What is fair compensation? (hourly rate, per-task, risk premium, royalties, other)
 - Should compensation differ based on emotional or moral labor?
3. Should people whose data, writing, art, or annotation labor helped develop AI systems share in the profits of AI products?
 - If so, how? Royalties, equity, licensing fees, or revenue..?

Be ready to share one insight with the class



One (*step towards a*) solution

Datasheets for datasets: Enabling transparency

Composition

What do the instances that comprise the dataset represent (e.g., documents, photos, people, countries)? Are there multiple types of instances (e.g., movies, users, and ratings; people and interactions between them; nodes and edges)? Please provide a description.

The instances are movie reviews extracted from newsgroup postings, together with a sentiment polarity rating for whether the text corresponds to a review with a rating that is either strongly positive (high number of stars) or strongly negative (low number of stars). The sentiment polarity rating is binary {positive, negative}. An example instance is shown in figure 1.

How many instances are there in total (of each type, if appropriate)?
There are 1,400 instances in total in the original (v1.x versions) and 2,000 instances in total in v2.0 (from 2014).

Collection Process

How was the data associated with each instance acquired? Was the data directly observable (e.g., raw text, movie ratings), reported by subjects (e.g., survey responses), or indirectly inferred/derived from other data (e.g., part-of-speech tags, model-based guesses for age or language)? Was the data reported by subjects or indirectly inferred/derived from data, was the data validated/verified? If so, please describe how.

The data was mostly observable as raw text, except the labels were extracted by the process described below. The data was collected by downloading reviews from the archive of the `rec.arts.movies.reviews` newsgroup at <http://reviews.imdb.com/Reviews>.

Motivation

For what purpose was the dataset created? Was there a specific task in mind? Was there a specific gap that needed to be filled? Please provide a description.

The dataset was created to enable research on predicting sentiment polarity—i.e., given a piece of English text, predict whether it has a positive or negative affect—or stance—toward its topic. The dataset was created intentionally with that task in mind, focusing on movie reviews as a place where affect/sentiment is frequently expressed.¹

Who created the dataset (e.g., which team, research group) and on behalf of which entity (e.g., company, institution, organization)?

The dataset was created by Bo Pang and Lillian Lee at Cornell University.

Distribution

Will the dataset be distributed to third parties outside of the entity (e.g., company, institution, organization) on behalf of which the dataset was created? If so, please provide a description.

Yes, the dataset is publicly available on the internet.

How will the dataset will be distributed (e.g., tarball on website, API, GitHub)? Does the dataset have a digital object identifier (DOI)?

The dataset is distributed on Bo Pang's webpage at Cornell: <http://www.cs.cornell.edu/people/pabo/movie-review-data>. The dataset does not have a DOI and there is no redundant archive.

When will the dataset be distributed?

The dataset was first released in 2002.

Conclusions

- Ethical failures in research were not anomalies; they reveal systematic risks when power and incentives are misaligned
- Informed consent, beneficence, and justice remain essential, but existing oversight mechanisms do not fully cover AI systems
- Modern AI relies heavily on human labor and human data, often without the protections afforded to research participants
- Historical cases help us see how harm can be normalized in the name of progress or innovation
- Ethical responsibility in AI and research requires accountability and being mindful of ethical principles that protect humans

Conclusions

Good practices

1. Pay annotators fairly, with compensation reflecting skill, time, and emotional labor; not minimum viable wages!
2. Acknowledge and compensate human contributors whose writing and feedback make AI systems possible
3. Engage with communities who generate or are represented in the data, not just extract from them
4. Build accountability beyond IRB compliance, including audits and community feedback
5. Transparency (datasheets for datasets)

References and recommended readings

Parts of this lecture were inspired by Yulia Tsvetkov's class (thank you!)

- [The Belmont Report.](#)
- Gebru, T., Morgenstern, J., Vecchione, B., Vaughan, J. W., Wallach, H., Iii, H. D., & Crawford, K. (2021). Datasheets for datasets. *Communications of the ACM*, 64(12), 86-92.

Feedback form—please use it!

<https://tinyurl.com/4sfsjszu>



EN 601.124

The Ethics of Artificial Intelligence and Automation:
Data: Human Subjects and Human Data

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Quizzes

- First quiz on Feb 9th
- Pen & paper, 15 minutes; testing basic understanding

Quizzes

Training a model to to **predict sexual orientation from images** faces which ethical challenges:

Select one:

- A) Reifying complex social constructs as stable categories
- B) Creating privacy risks by exposing confidential information
- C) Using people's data for purposes they never consented to
- D) All of the above

Quizzes

Which statement is true about **generative text-to-image models**:

Select one:

- A) They have no ethical risks as they are not used for predictions
- B) They can amplify biases present in training data
- C) They generate structured predictions, scores, or categories
- D) All of the above